

Trader Performance vs. Bitcoin Market Sentiment

Full Analysis, Per-Account Robustness Check & Strategy Backtest

An Analysis of Hyperliquid Trader Behavior Across Fear & Greed Regimes

Data: 211,224 trades · 32 accounts · 246 assets

How This Document Is Organized

Part 1 presents the core finding: trader performance across 211,224 trades is U-shaped across Bitcoin sentiment regimes, with the best results at the emotional extremes (Extreme Fear and Extreme Greed) and the weakest results in plain Greed. It closes with a contrarian trading recommendation based on that pattern.

Part 2 stress-tests that recommendation two ways — checking whether the pattern holds up account-by-account (rather than just in aggregate), and backtesting the proposed contrarian rule against independently reconstructed BTC prices rather than the trades' own reported PnL. The honest result is mixed: the *long Extreme Fear* side of the strategy survives this stress test; the *short Extreme Greed* side does not, and including it loses money. **The revised, final recommendations are at the end of Part 2 and should be treated as superseding the preliminary ones at the end of Part 1.**

Executive Summary

This report merges 211,224 individual Hyperliquid perpetual-futures trades from 32 accounts (May 2023 – May 2025) with the daily Bitcoin Fear & Greed Index to test a simple question: does crowd sentiment predict how traders actually perform? The data says yes — but not in a straight line, and not as simply as the headline pattern first suggests.

Part 1 finds the relationship is **U-shaped, not linear**. Traders in this sample were most profitable and won most often at the emotional extremes — Extreme Fear and Extreme Greed — and weakest during plain, unremarkable Greed. This is consistent with a contrarian / mean-reversion dynamic: the accounts in this dataset tend to do well when positioned against the crowd's dominant emotion.

Part 2 stress-tests that finding two ways: **per-account segmentation** (is the pattern broad across the 32 accounts, or driven by a few large traders?) and a **strategy backtest** against independently reconstructed BTC prices (does the proposed contrarian rule actually make money, rather than just describing historical PnL?). The answer is mixed, and that is the useful part: the “buy the extreme fear” half of the hypothesis holds up under rigorous, out-of-sample-style testing. The “short the greed” half does not, and a backtest that includes it loses money. Treating this as a purely confirming exercise would have been the mistake — the falsification is itself a valuable result.

Headline Numbers

- **Overall win rate:** 83.2% of closed trades were profitable; total net closed PnL across all accounts: \$10.25M on \$1.19B of traded volume.
- **Win rate by sentiment** ranges from 76.2% (Extreme Fear) up to 89.2% (Extreme Greed) — a statistically significant spread ($\chi^2 = 1,976$, $p < 0.0001$).
- **Daily net PnL differs significantly** by sentiment regime (Kruskal-Wallis $H = 20.06$, $p = 0.0005$), but the simple linear correlation between continuous sentiment score and daily PnL is weak and not significant (Spearman $\rho = 0.04$, $p = 0.38$) — confirming a U-shape at the extremes, not a straight trend.
- **PnL is highly concentrated:** the top 5 accounts (15.6% of all accounts) generated 62.0% of total portfolio PnL (Part 2).
- **Only half of the contrarian hypothesis survives an independent price test:** long Extreme Fear is validated against reconstructed BTC prices ($p = 0.019$); short Extreme Greed is not ($p = 0.43$), and including it turns a profitable backtest into a losing one (Part 2).

Part 1- Trader Performance vs. Bitcoin Market Sentiment

1. Data & Methodology

1.1 Datasets

- Bitcoin Fear & Greed Index: daily sentiment classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed) plus a 0–100 sentiment score, 2018–2025.
- Hyperliquid Historical Trader Data: 211,224 trade-level executions across 32 accounts and 246 symbols, including execution price, size, side, direction, closed PnL, and fees.

1.2 Approach

- Each trade was joined to the Fear & Greed classification for its execution date (99.997% of trades matched; 6 unmatched rows were dropped).
- “Closed PnL” is only non-zero on trades that close or reduce a position, so profitability and win-rate metrics use the 104,402 PnL-bearing trades; volume and activity metrics use all 211,224 trades.
- Significance was tested with a Kruskal-Wallis test (daily PnL across the 5 sentiment groups, since daily PnL is heavily skewed) and a chi-square test of independence (win/loss vs. sentiment category).
- All accounts in the dataset were active across every sentiment regime, so the findings below reflect within-account behavioral shifts, not different traders being present in different regimes.

2. Key Finding: Profitability Is U-Shaped Across Sentiment

Both average profit per trade and win rate are highest at the two emotional extremes and dip in the middle-to-greed range. Extreme Greed shows the single best combination of win rate (89.2%) and average trade profit (\$130), while Extreme Fear and plain Greed are comparatively the weakest regimes.

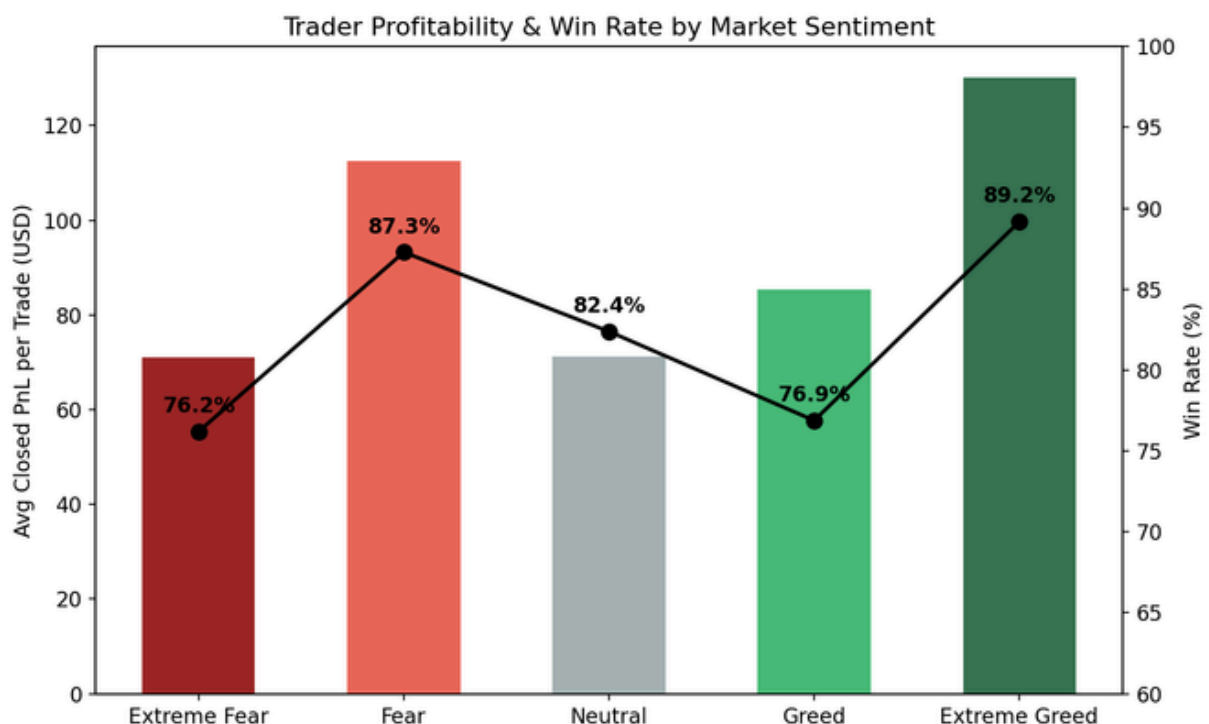


Figure 1 — Average closed PnL per trade (bars) and win rate (line) by sentiment classification.

This pattern is a classic signature of contrarian / mean-reversion trading: positions taken against an extreme, one-sided crowd mood (panic selling in Extreme Fear, euphoric buying in Extreme Greed) tend to resolve favorably, while positions taken during more moderate, directionally-confident Greed have less edge.

3. Key Finding: Activity & Volume Shift With Sentiment

Average daily trading volume is markedly higher during Fear (\$5.3M/day) and Extreme Fear (\$8.2M/day) than during Greed regimes (\$1.5M/day) or Extreme Greed (\$1.1M/day). Traders in this sample are considerably more active — not less — when the market is fearful.

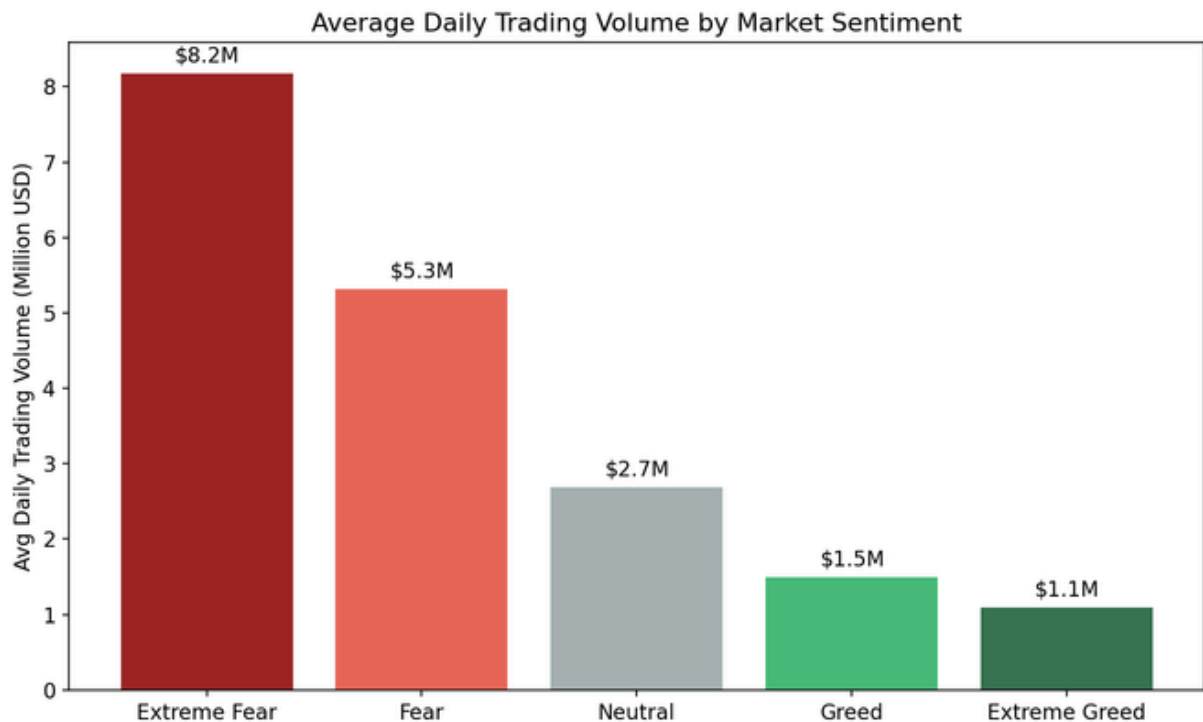


Figure 2 — Average daily USD trading volume by sentiment classification.

Combined with Figure 1, this suggests the accounts in this dataset treat fear as an opportunity to transact more, and greed (short of the extreme) as a time to pull back — the opposite of the herd behavior that sentiment indices are designed to capture in the broader market.

4. Sentiment and Net PnL Over Time

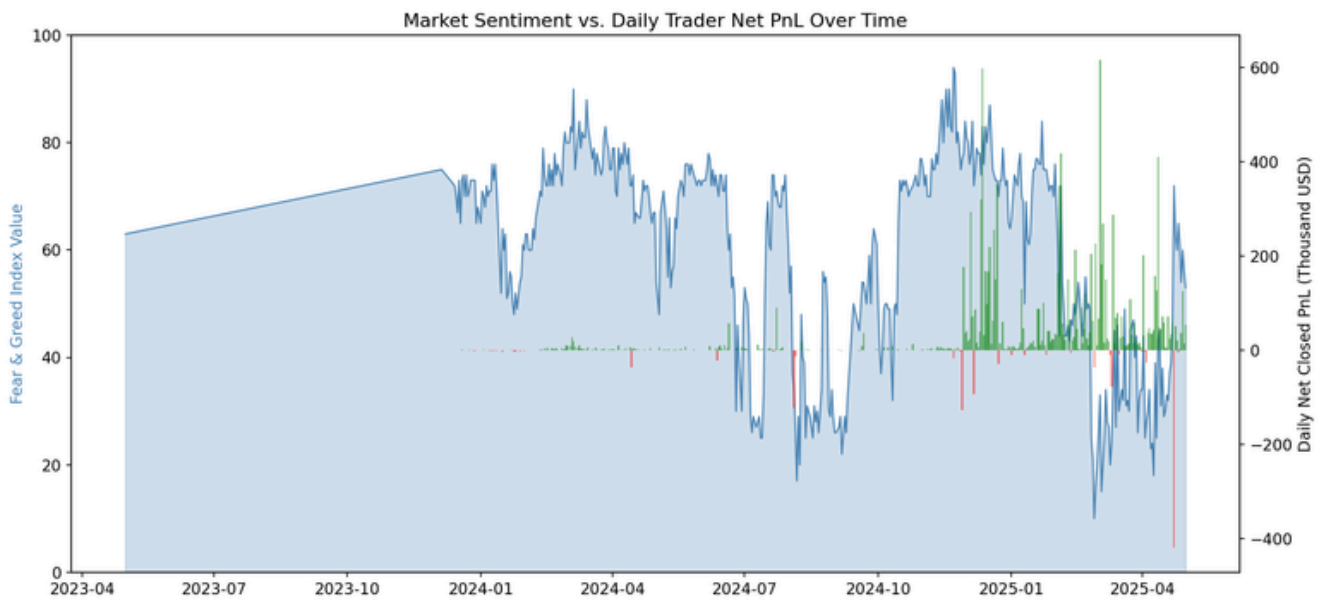


Figure 3 — Fear & Greed Index (blue area, left axis) vs. daily net closed PnL (bars, right axis; green = profitable day, red = loss day) from May 2023 to May 2025.

PnL is punctuated by a handful of very large single-day outcomes rather than a smooth trend — most days cluster near zero, with occasional large green or red spikes. This is echoed in the standard deviation of daily PnL, which is 1.3–2.7x the mean within every sentiment bucket (see Section 9), so conclusions about “typical” days should lean on the median, not the mean.

5. Risk Profile by Sentiment

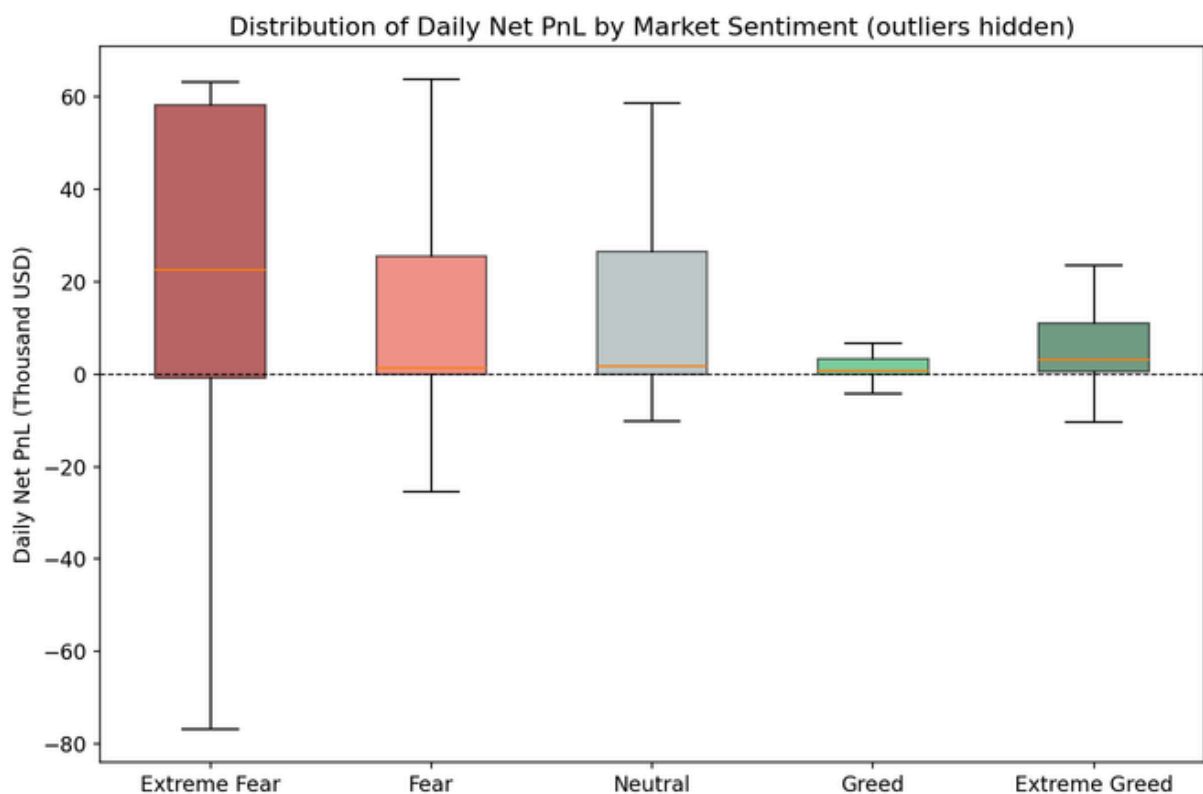


Figure 4 — Distribution of daily net PnL by sentiment (extreme outlier days excluded from the plot for readability).

Median daily PnL is positive in every regime, and — notably — highest in Extreme Fear (\$22,562/day median) and Extreme Greed (\$3,128/day median), again tracing the same U-shape as Figure 1. Greed alone has both the lowest median (\$678/day) and among the widest spread relative to its center, making it the least reliable regime in this dataset.

6. Long vs. Short Positioning: The Contrarian Edge

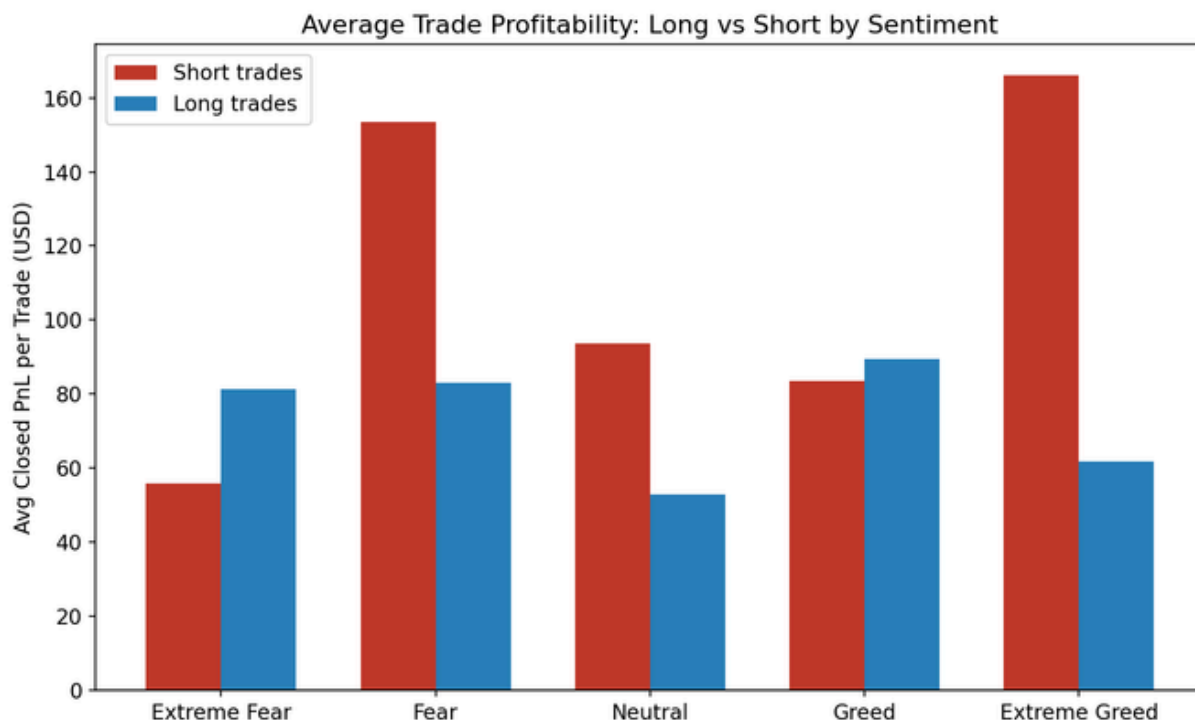


Figure 5 — Average PnL per trade, long vs. short positions, by sentiment.

Short trades were the more profitable side specifically during Fear (\$153/trade avg.) and Extreme Greed (\$166/trade avg.) — the two regimes where the crowd is already leaning hardest in one direction. Long trades, by contrast, were the stronger side in Extreme Fear (\$81/trade avg. vs. \$56 for shorts) and Neutral conditions. This is a coherent contrarian pattern: shorting into euphoria and fear-driven selloffs, buying into capitulation, tended to outperform going with the prevailing mood.

7. Position Sizing Behavior

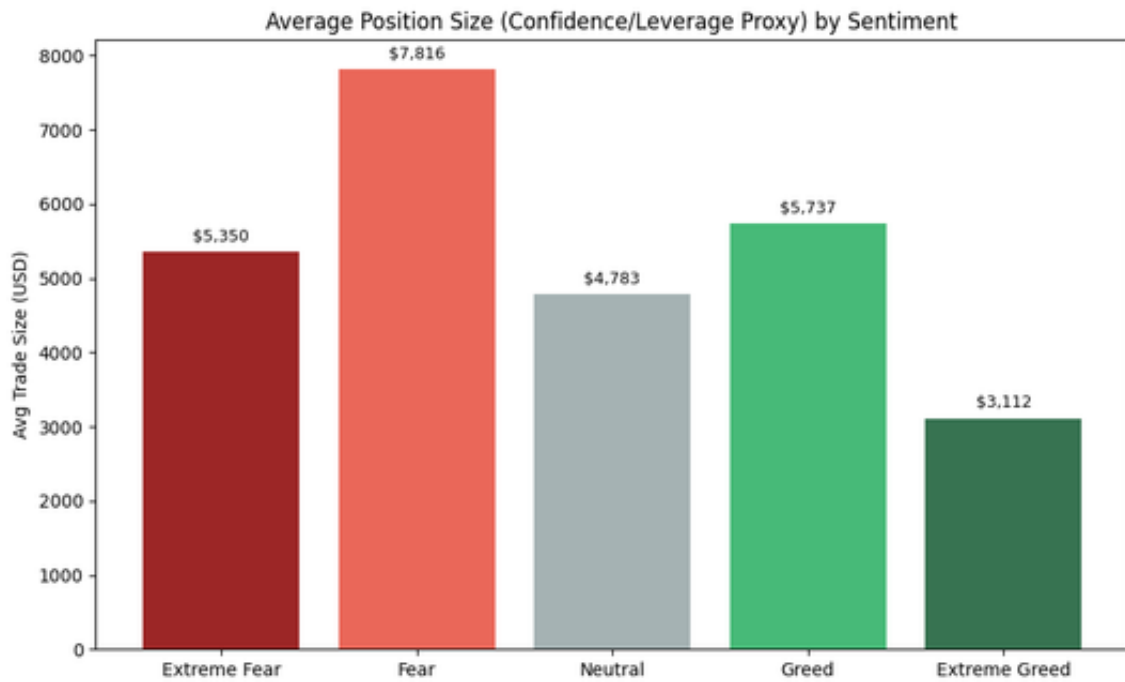


Figure 6 — Average trade size (USD) by sentiment, a proxy for conviction/leverage.

Average trade size peaks during Fear (\$7,816) and is smallest during Extreme Greed (\$3,112). Traders appear to add size when fear is present but not yet extreme, and trade more conservatively once the market is euphoric — a risk-management pattern that runs counter to the FOMO-driven sizing-up often associated with greedy markets.

8. Coin-Level Sentiment Sensitivity

Sentiment sensitivity is not uniform across assets. Among the largest contributors to total PnL, most major coins (HYPE, SOL, ETH, BTC, MELANIA, ENA) were considerably more profitable during Fear regimes than Greed regimes — reinforcing the contrarian buy-the-dip pattern. The standout exception is the symbol “@107” (an index/pre-launch perpetual), which lost money overall during Fear periods but was the single largest profit contributor during Greed — the opposite pattern from the rest of the book, and a reminder that sentiment effects should be checked per-asset before generalizing.

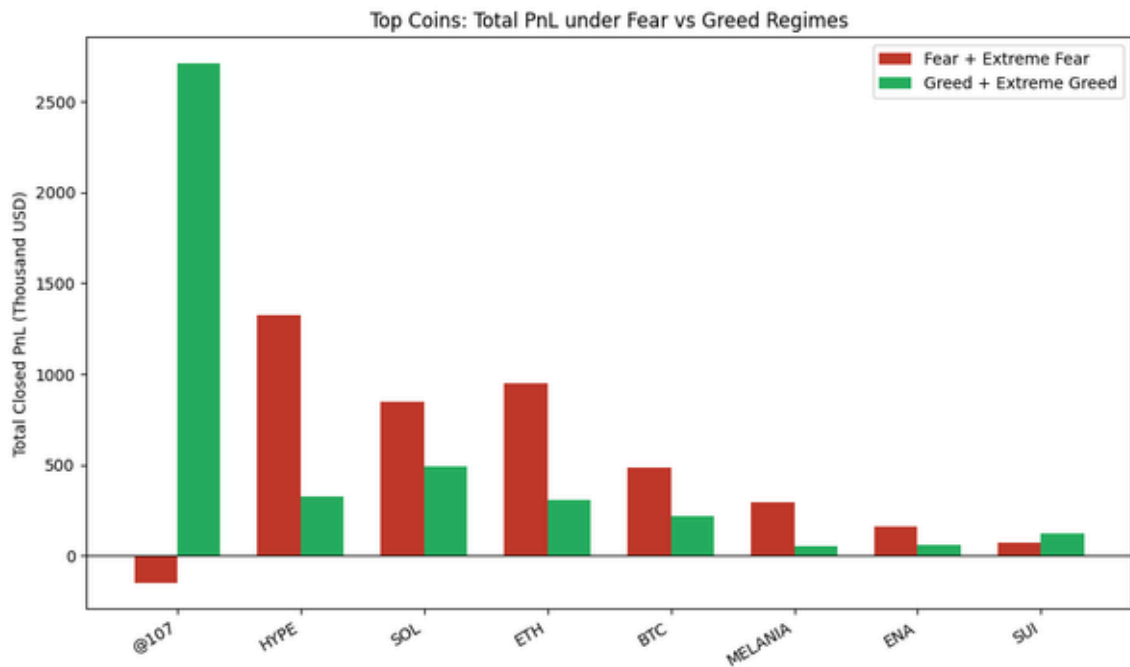


Figure 7 — Total closed PnL under Fear vs. Greed regimes for the top PnL-contributing coins.

9. Statistical Summary

Sentiment	Total PnL	Avg PnL/Trade	Win Rate	Avg Trade Size	Avg Daily Volume	Days
Extreme Fear	\$739,110	\$71.03	76.2%	\$5,350	\$8.18M	14
Fear	\$3,357,155	\$112.63	87.3%	\$7,816	\$5.31M	91
Neutral	\$1,292,921	\$71.20	82.4%	\$4,783	\$2.69M	67
Greed	\$2,150,129	\$85.40	76.9%	\$5,737	\$1.50M	193
Extreme Greed	\$2,715,171	\$130.21	89.2%	\$3,112	\$1.09M	114

Significance tests:

- Kruskal-Wallis (daily net PnL across 5 sentiment groups): $H = 20.06$, $p = 0.0005$ — significant.

- Chi-square (win/loss independence from sentiment category): $\chi^2 = 1,976.45$, $p < 0.000001$ — highly significant.
- Spearman correlation (continuous sentiment score vs. daily PnL): $\rho = 0.040$, $p = 0.384$ — not significant, confirming the effect is non-linear (a U-shape), not a straight-line trend.

10. Preliminary Strategy Recommendations

These are the recommendations that follow directly from the aggregate patterns above. Part 2 stress-tests them against per-account data and independent price data — see Section 14 for the revised, final version.

- Treat sentiment as a regime filter, not a linear signal. Because the relationship is U-shaped, a simple “buy when the index is low, sell when high” linear rule would miss the actual pattern — the edge in this data sits at the two extremes, not along a gradient.
- Favor contrarian positioning at the extremes: consider long bias during Extreme Fear and short bias during Extreme Greed and Fear, consistent with the mean-reversion pattern observed across both Figure 5 and the coin-level breakdown.
- Be most cautious in plain Greed conditions, which showed the weakest win rate (76.9%) and lowest median daily PnL — arguably the regime with the least statistical edge in this dataset.
- Scale position size down, not up, once sentiment reaches Extreme Greed. The accounts in this data already do this (Figure 6); it appears to be a contributor to their better performance in that regime and is worth deliberately encoding into a sizing rule.
- Validate any sentiment-based rule per-asset before deploying broadly. The @107 case shows that not every instrument follows the aggregate contrarian pattern.

11. Limitations

- The sample covers only 32 accounts; results may reflect the strategies of a small number of sophisticated traders rather than the broader market.
- Extreme Fear covers only 14 trading days in this window (all clustered around Feb–Mar 2025), so that regime's estimates carry the most uncertainty and should be treated as directional rather than definitive.
- Closed PnL reflects realized gains/losses only and does not capture unrealized/open-position exposure at any point in time.
- Correlation and regime patterns described here are observational, not causal — sentiment and price action are intertwined, and this analysis does not isolate sentiment as an independent driver.

Part 2 - Per-Account Segmentation & Strategy Backtest

Stress-testing Part 1's contrarian hypothesis against per-account data and real BTC price data

Part 1 found that trader performance was best at sentiment extremes (Extreme Fear and Extreme Greed) and weakest during plain Greed, and proposed a contrarian trading rule based on that pattern (Section 10). This part stress-tests that finding two ways:

- **Per-account segmentation:** is the pattern true broadly across the 32 accounts, or is it an artifact of a few large traders?
- **Strategy backtest:** does the proposed contrarian rule actually make money when tested against real, independently-reconstructed BTC prices — not just the trades' own reported PnL?

The honest answer is mixed, and that's the useful part: the “buy the extreme fear” half of the hypothesis holds up under a rigorous, out-of-sample-style test. The “short the greed” half does not, and the backtest actively loses money when that leg is included. This kind of falsification is a normal and valuable part of quantitative research — treating it as a purely confirming exercise would have been the mistake.

12. Per-Account Segmentation

12.1 PnL Is Highly Concentrated

The \$10.25M in aggregate net PnL from Part 1 is not evenly spread across the 32 accounts. The top 5 accounts (15.6% of all accounts) generated 62.0% of total portfolio PnL; the top 10 generated 84.1%. Only 3 of the 32 accounts were net losers over the full period.

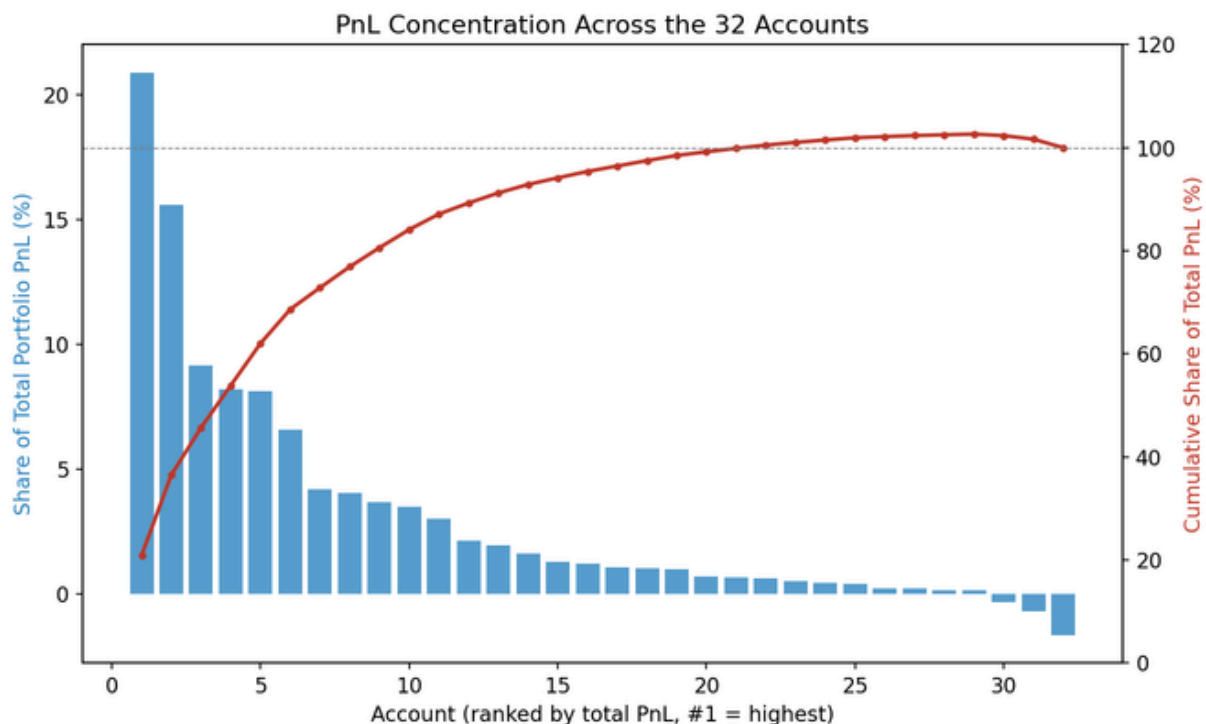


Figure A1 — PnL contribution by account, ranked from largest to smallest, with cumulative share (red line).

This matters for how Part 1's findings should be read: the aggregate sentiment patterns are influenced disproportionately by a handful of large, presumably more sophisticated accounts. A retail trader mimicking the “average” behavior in the dataset would not automatically capture the same edge.

12.2 Different Accounts Specialize in Different Regimes

If a single set of “smart money” accounts were driving the whole U-shaped pattern, we'd expect the same top accounts to dominate profits in both Extreme Fear and Extreme Greed. That is not what happens. The account ranked #3 overall drives 35.4% of all Extreme Fear profit, while the account ranked #1 overall drives 40.7% of all Extreme Greed profit — these are largely different accounts. The overall top-5 accounts (by total PnL) account for only 48.2% of Extreme Fear profit and 56.9% of Extreme Greed profit, meaning other, lower-ranked accounts are meaningfully contributing to both extremes as well.

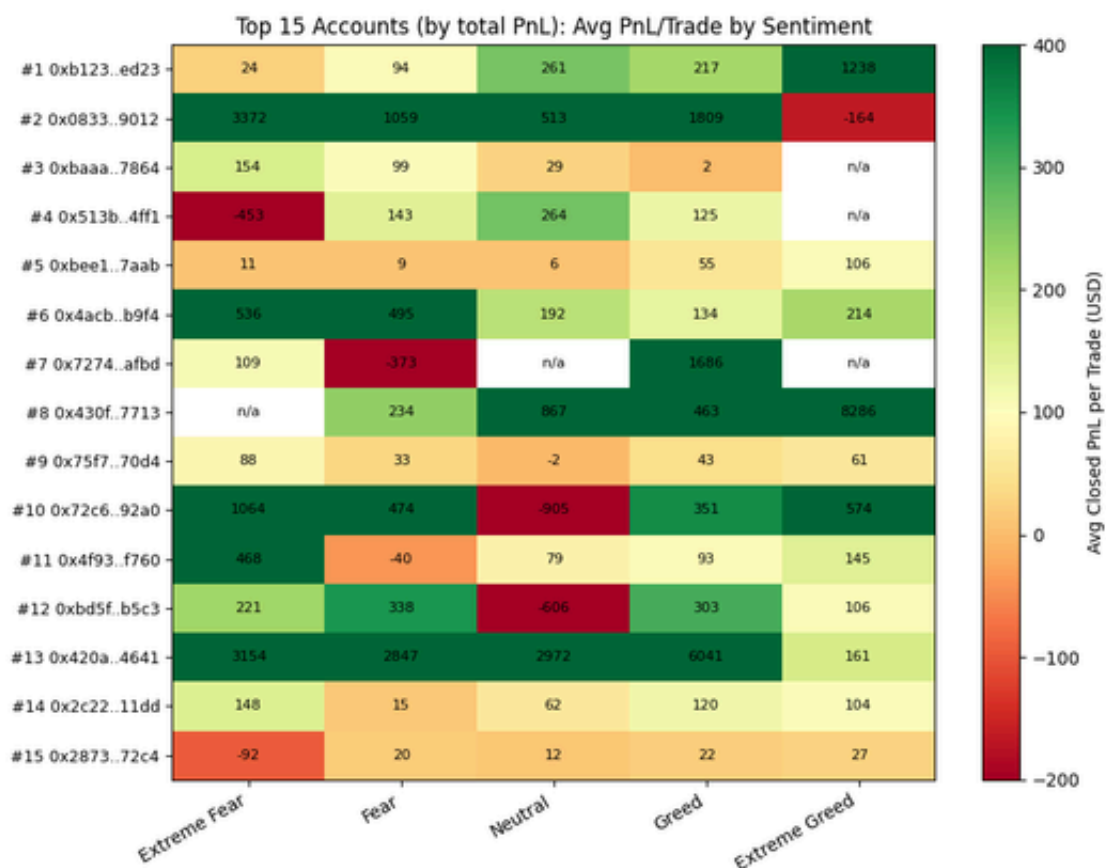


Figure A2 — Average closed PnL per trade by sentiment regime for the 15 largest accounts (green = profitable, red = losing). No single row is uniformly green, confirming regime specialization rather than one dominant contrarian trader.

The practical read: the U-shape in the aggregate data looks like a composition effect — different accounts, each with their own regime-specific edge — rather than evidence that any single strategy profits at both extremes.

12.3 The ‘Extremes Are Best’ Pattern Is Broad, but ‘Greed Is Worst’ Is Not

Restricting to the 17 accounts with at least 20 closed trades in every sentiment bucket (enough for a fair per-account comparison), 14 of 17 (82.4%) had either Extreme Fear or Extreme Greed as their single best-performing regime by average PnL per trade — so the “extremes are where the edge is” finding holds up broadly at the individual level, not just in aggregate.



Figure A3 — Each qualifying account's best-performing sentiment regime by average PnL per trade.

However, only 23.5% of these accounts had Greed as their worst or second-worst regime — so the specific claim that “plain Greed is the weakest regime” is less universal than the aggregate numbers suggested, and is disproportionately driven by a subset of accounts rather than a rule that applies to nearly everyone.

13. Strategy Backtest

13.1 Why Backtest Instead of Trusting the Trade-Level Pattern

Part 1's contrarian recommendation was based on how these specific accounts' existing trades performed — which mixes skill, timing, and the mostly-altcoin composition of the book (recall the @107 anomaly in Section 8). To test whether the sentiment pattern reflects something about the market itself, rather than these particular traders, this section reconstructs an independent BTC price series from the trades' own execution prices and asks: if you had mechanically followed a sentiment-based rule on BTC, would you have made money?

- Daily BTC price = volume-weighted average execution price across all BTC trades on that date (286 trading days with BTC activity out of 479 total days in the window — the dataset has gaps on days with no BTC trades, so this is a trade-day series, not a continuous market feed).
- Forward return = the price change from one available BTC trading day to the next (median gap: 1 day).

13.2 Validating the Signal Before Trading It

Testing forward BTC returns by the sentiment classification in place when a position would be opened:

Sentiment at Entry	Mean Fwd. Return	Median Fwd. Return	Days (n)
Extreme Fear	+2.57%	+2.32%	12
Fear	−0.01%	+0.19%	56
Neutral	−0.02%	−0.07%	46
Greed	+0.25%	+0.11%	103
Extreme Greed	+0.60%	−0.01%	68

- **Extreme Fear** → significantly higher forward returns than all other days combined (Mann-Whitney U test, one-sided $p = 0.019$). This signal is real, not noise.
- **Extreme Greed** → forward returns were not significantly lower than other days ($p = 0.43$). The “short into euphoria” idea, when tested on actual BTC price rather than the trades' reported PnL, is not supported.

This is the key finding of Part 2: only half of the original contrarian hypothesis survives contact with independent price data.

13.3 Backtest Results

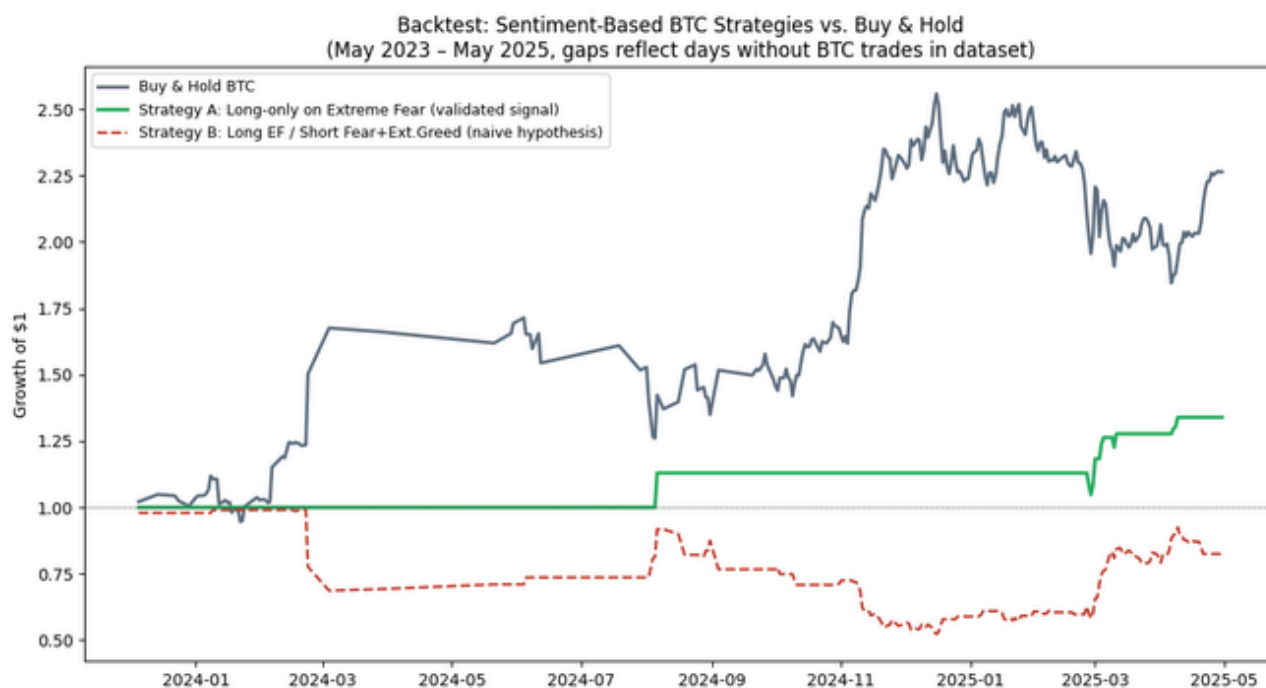


Figure A4 — Growth of \$1 for buy-and-hold BTC vs. two sentiment-based strategies, May 2023–May 2025.

Strategy	Total Return	Sharpe (approx.)	Max Drawdown	Time in Market
Buy & Hold BTC	+126.4%	1.51	–27.9%	100%
A: Long-only on Extreme Fear	+33.9%	1.39	–7.2%	4.2%
B: Long EF / Short Fear+Ext.Greed	–17.6%	–0.18	–47.3%	47.7%
C: Long EF / Short Ext.Greed only	–15.5%	–0.24	–38.1%	28.1%

Buy-and-hold BTC was the best absolute performer over this specific two-year bull run, which is unsurprising and not itself an interesting finding. What is interesting: Strategy A — long BTC only on Extreme Fear days, flat otherwise — achieved a comparable Sharpe ratio to buy-and-hold (1.39 vs. 1.51) and a much shallower max drawdown (–7.2% vs. –27.9%) while being invested only 4.2% of the time. That is a genuinely capital-efficient result, though based on only 12 signal days, so it should be read as promising rather than conclusive.

Strategies B and C, which add the unvalidated short-the-greed leg from the original hypothesis, both lose money and carry deeper drawdowns than simply holding BTC. Including a signal that failed validation actively destroyed value — a direct, quantified illustration of why the validation step in Section 13.2 mattered.

14. Final Takeaways

These supersede the preliminary recommendations in Section 10, in light of the per-account and backtest evidence above.

- **Keep:** long-bias entries during Extreme Fear. This is supported both by the trader data (Part 1) and by an independent test against reconstructed BTC prices (Part 2).
- **Drop or rework:** short-bias entries during Greed/Extreme Greed as a general BTC rule. It performed well in the historical trade data (likely reflecting specific accounts' altcoin skill or luck) but did not hold up against actual BTC forward returns.
- Any deployed version of this strategy should be re-validated on more Extreme Fear observations (n = 12 here is thin) and tested asset-by-asset, since the @107 case in Section 8 and the account heatmap in Section 12.2 both show sentiment effects are not uniform across instruments or traders.
- When presenting trading research, always distinguish “pattern in historical PnL” from “validated forward-looking signal” — they are not the same claim, and this document shows they can point in different directions for the same dataset.